

Capacity Eviction in Reconstruction-Trained Tokenizers

Draft of August 10, 2026 — section text frozen pending review

Note to the reviewer. This is §4 of a paper in preparation, exported verbatim from the manuscript with its two figures. Numbering matches the manuscript: §4.1–§4.8, Equations (1) and (2), and Figures 5 and 6. §4 specifies the experiment; it reports no result, and the pre-registered run has not been executed. References to §3, §7 and Figure 8 point outside this export and are bound to their manuscript numbers.

What we would most like challenged. (i) Whether the five cells, as specified, can separate microstructure *information* from input *capacity* — §4.3 states that they cannot fully, and quantifies nothing; we would like to know whether the disclosure is sufficient or whether the design should not ship without the sixth cell. (ii) Whether the execution filter and its selection procedure (§4.4) are sound given that the filter never bound on the calibration fixture. (iii) Whether the decision rule (§4.6) is genuinely falsifiable — whether there is a result we would be forced to call a null. (iv) Whether the multiple-testing and power treatment (§4.7) is honest about what five seeds can and cannot resolve. (v) Any number that does not follow from what is stated.

4 Experimental design

The mechanism of Section 3 says that a reconstruction-trained tokenizer will evict microstructure unless it is built not to. This section describes the experiment that asks the separate question the paper is organized around: whether real trade-flow imbalance, carried through a tokenizer built to preserve it, survives to a cost-aware trading decision. The design was frozen before any of its numbers existed, and this section describes it as frozen.

4.1 Five cells

The experiment is a 2×2 over the two factors that could plausibly explain a difference — the quantizer and the input arm — plus a fifth cell that isolates the one thing a 2×2 cannot: whether an effect attributed to microstructure is really microstructure. Figure 5 shows the layout; the registry that the runner reads is the authority.

Cells 1 and 2 receive the seven OHLC-shape, volume and amount dimensions only; cells 3 and 4 additionally receive the six microstructure dimensions (indices 7–12: trade-flow imbalance, signed count imbalance, trade count, mean trade size, trade-size dispersion, large-trade share). Cells 1 and 3 quantize with binary spherical quantization; cells 2 and 4 with finite scalar quantization. Cell 5 is cell 4 with the microstructure block shuffled, and is the placebo.

The comparison that carries the paper is the paired difference $\Delta IR(4 - 5)$. The others are support: $IR(4) - IR(2)$ asks whether the microstructure arm beats the OHLCV arm without reference to the placebo, and the two BSQ cells anchor the quantizer axis. We are explicit in Section 7 that the three marginals involving a BSQ arm are withdrawn as claims, because the BSQ implementation has no external reference point; they are reported, not claimed.

In the OHLCV-only arm the six microstructure dimensions are *removed* from the input rather than supplied and left uninformative, so that arm's feature width is seven rather than sixteen. That choice matters for interpreting the legibility gate, which is undefined for an arm with no microstructure channel and is exempted there by a named registry entry rather than by silence.

4.2 Capacity is controlled, not assumed

A comparison between two quantizers is meaningless if one of them simply has more room. We therefore control the two quantities that determine room, and report both rather than asserting parity. The finite-scalar arm uses per-dimension levels (11, 9, 9, 7, 7, 5, 5), split into a coarse subtoken of 891 codes and a fine subtoken of 1225, for 20.058 bits per bar; the binary-spherical arm spends $10 + 10$ bits for 20.000. The two differ by 0.058 bits per bar, inside a pre-registered parity band of 19.5–20.5.

The backbone realizes 21,301,248 parameters at the finite-scalar vocabulary and 21,231,616 at the binary-spherical one — a difference of 69,632 parameters, or 0.327%, arising entirely from vocabulary-dependent embedding and output-head rows. The backbone is otherwise identical across all five cells: eight layers, model width 512, feed-forward width 1024, eight heads, and a context of 512 bars. It is held fixed on purpose. The tokenizer is the object under study; the backbone is the instrument that measures it.

Every cell trains under the same budget: 26,003 steps at batch 32 and context 512, on the same draw. That budget is matched and fixed rather than chosen per cell by early stopping. The alternative — stopping each cell when its own validation loss saturates — would give five different budgets and confound $\Delta\text{IR}(4-5)$ with “cell 4 trained longer”, which is the same class of confound the placebo exists to remove. The cost is that no cell is guaranteed to be at its own optimum; saturation is measured and reported rather than optimized against.

4.3 The placebo, and the handicap it creates

The placebo answers a question no capacity control can: whether an effect attributed to microstructure comes from microstructure’s *information* or merely from the model having six more input channels to spend capacity on. Cell 5 receives the same six channels, with the same marginal distributions, carrying no usable information.

The shuffle permutes the six microstructure dimensions as a block, within each contiguous segment of a symbol, with a full within-segment permutation rather than a fixed block length. Three properties are enforced rather than assumed. The block moves as a unit, so within-bar structure *among* the microstructure dimensions survives and only their relationship to everything else is destroyed. The mask travels with the value, because permuting values alone would strand fill values on valid bars and real values on masked ones — a defect found in review and fixed before any training. And the permutation is derived from the run seed and the symbol together, so the train and evaluation paths shuffle identically.

Figure 5 reports what the shuffle measurably does. It destroys the contemporaneous link between microstructure and price shape, from mean absolute correlation 0.2037 to 0.0005, and it destroys microstructure’s own lag-1 autocorrelation, from 0.4165 to 0.0006. It leaves the within-block correlation at 0.2886 and the within-OHLCV correlation at 0.3105, both unchanged.

This is where the design’s principal known weakness lives, and we state it rather than let a reader find it. Because cell 4’s microstructure dimensions are correlated with price at 0.20 and autocorrelated at 0.42, part of the capacity they consume is *amortized* against structure the code is already carrying. Shuffling removes that amortization along with the information, so cell 5 pays a capacity price cell 4 does not. The consequence is a forfeit stated in the pre-registered rule string itself: $\Delta\text{IR}(4-5)$ is (information) + (capacity handicap), and the paper does not claim the information effect alone clears any threshold. The handicap is not quantified in information-ratio units, and Section 7 carries that as an open disclosure rather than a resolved question.

4.4 The headline metric

The headline is a cost-aware net information ratio, annualized, computed on a portfolio return series rather than on pooled per-position returns. For a position of sign s opened at the close of bar $t + 1$ and closed h bars later, the realized net log-return subtracts execution and funding cost directly:

$$r^{\text{net}} = s \log \frac{C_{t+h}}{C_{t+1}} - c_{\text{total}} - c_{\text{funding}}, \quad c_{\text{total}} = 2 c_{\text{side}}. \quad (1)$$

Each period’s portfolio return is the equal-weighted mean over the symbols active in that period, and the information ratio is computed on that series, so a symbol that is not trading contributes nothing rather than contributing a zero.

The per-side cost has three additive components — a taker fee, a volatility-scaled half-spread floored at the symbol’s own decile spread, and a participation-based impact term:

$$c_{\text{side}} = f_{\text{taker}} + \frac{1}{2} \text{spread} \cdot \max(1, \sigma_t / \bar{\sigma}) + k_{\text{impact}} (Q / \text{ADV}). \quad (2)$$

This model is a secondary diagnostic. **The headline nets a flat 0.30% round trip**, the pessimistic end of the pre-registered 0.1–0.3% band, pinned on the conformance surface and hard-checked before

any cell is scored. Reporting the headline at the worst end of the band, and the model-based figure alongside it, is deliberate: a result that needs the optimistic end of its own cost assumption is not a result.

Positions are taken only when the forecast is large enough to be worth its cost. The execution filter is $\theta = \kappa c$: trade only when $|\hat{\mu}| > \theta$, with κ selected on the validation block from the grid $\{1.0, 1.5, 2.0, 3.0\}$ and then held fixed.

One measured property of this filter is worth stating, because it changes how a calibration result should be read. At the pinned horizon the threshold never bound on the fixture’s degenerate checkpoints: for the three noise-arm cells re-scored there, the set of periods traded equalled the set scored — 12,000 of 12,000 — at every κ and at both volatility scalings tested, which makes the realized information ratio on those cells a pure function of $\text{sign}(\hat{\mu})$ rather than of forecast magnitude. At shorter horizons the same checkpoints fail at the opposite endpoint, trading nothing at all: one of the three traded zero of 12,000 at $h \in \{1, 2, 3, 5\}$, and every noise checkpoint we hold is pinned to one endpoint or the other at $h \leq 3$. Both are the same pathology — a filter that never discriminates — and the guard below treats them alike. On a non-degenerate checkpoint of the *same* fixture it bound at every κ : the planted cell 4 traded 96.2%, 94.4%, 92.5% and 88.7% of scored decisions at $\kappa = 1.0, 1.5, 2.0, 3.0$. The two readings together identify the cause, and it is not the fixture. Binding is decided by the *forecast* magnitude, never by the realized return: on the non-binding cells $\hat{\mu}$ had collapsed to a near-constant per-step offset — a decode artifact, with $|\text{mean}|$ between 23 and 283 times its own standard deviation across the three cells — so at $\kappa = 1.0$ it sat 4.4 to 31.6 thresholds away from zero while its own standard deviation was only 0.11 to 0.48 of one threshold, and cleared the filter everywhere. Raising the volatility scaling moved those cells *further* from binding rather than nearer, which is the signature of an offset rather than of a threshold set too low.

Whether the filter binds on real data is therefore an open question about the trained models, not about the design, and we do not assume the answer. Nothing in this repository measures it: no eval artifact from a real-lake run retains a decision-activity value, so the question is genuinely unanswered rather than answered elsewhere. What the pre-registration commits to instead is a refusal. The degeneracy guard reads each cell’s decision-activity at κ^* and its fraction of negative forecasts, and returns `HALT_ADJUDICATE` — not a null — for any cell whose book is constant-direction. We record one asymmetry in that guard rather than discovering it later: the sign-balance leg carries a band, halting outside $[0.05, 0.95]$, while the filter-binding leg tests the exact endpoints only, so an activity of 0.999 passes where a fraction-negative of 0.999 halts. A near-degenerate book on the activity leg would not be caught, and the project has already measured a 0.99883 reading on the sibling leg, so the case is not hypothetical.

Forecasts are the conditional mean $\hat{\mu}$, not the mode. The estimator was changed from greedy-argmax decoding to an expectation decode after argmax was found to compound a sign bias over a rollout — fixture mean -0.051 against a standard deviation of 0.036, with frac-negative 0.929, versus -0.004 and 0.534 under expectation, while the correlation between $\hat{\mu}$ and the planted state *rose* from 0.858 to 0.898. The change is a correctness fix, made before any real cell was scored, and it re-anchored the reproducibility hash because the predictor sits inside the anchored evaluation instrument.

Information coefficient, rank information coefficient, mean absolute error and R^2 are computed and reported as secondary diagnostics. They do not enter the decision rule.

4.5 Evaluation geometry

Models are trained once and evaluated forward; nothing is refit per fold. Figure 6 shows the geometry. The four-year window is split at its 0.7 point, on 2023-10-20, into a training region and a forward region; the forward region is cut into six equal blocks. Block 0 is spent selecting κ and is excluded from every headline number, so the headline is scored on blocks 1–5, spanning 2024-01-01 to 2025-01-01 — a region the selection never saw.

Between any training window and a label that could contaminate it sit two margins: a 60-bar label look-forward, which is the maximum horizon the autoregressive rollout reaches, and a 60-bar serial-correlation margin. Their sum is a 120-bar leading embargo.

The premise behind the second margin — that 60 bars is enough for serial correlation to have

decayed — is measured rather than assumed, on all 200 instruments, with a control arm that recovers a known AR(1) answer first so that a null reading is a measurement and not a silent failure. For signed returns, the channel through which a label leak would actually travel, the autocorrelation at lag 60 is 0.0067 on average and 0.0191 at the worst symbol, against a margin of 60: roughly a 24× margin. For absolute returns it is 0.2323 on average and 0.3604 at the worst symbol, and the premise does *not* hold. That is volatility clustering rather than a label-leakage channel, and Section 7 states what is and is not established by measuring only the signed channel.

Confidence intervals on ΔIR come from a paired moving-block bootstrap over the portfolio series, $B = 10,000$ resamples at a fixed seed, with $\lceil \sqrt{T} \rceil$ blocks and percentile intervals. The pairing is not cosmetic. Cells 4 and 5 hold the same market exposure, so the common factor cancels in the paired series; the average raw cross-sectional correlation is 0.6387, which leaves an effective breadth of about 1.5 symbols out of 40. The cross-section here buys robustness, not statistical power, and we say so rather than counting 40 independent bets.

4.6 The decision rule

The rule was written, timestamped and committed before any cell was trained. Figure 8 shows it as the runner evaluates it: five clauses, conjunctive, so the primary outcome is SURVIVES only if none fails and NULL otherwise.

1. **Paired interval.** The one-sided 95% lower bound of $\Delta\text{IR}(4 - 5)$ exceeds zero.
2. **Detectability.** $\Delta\text{IR}(4 - 5)$ reaches the paired minimum detectable effect, so an effect too small for this design to resolve cannot be reported as one.
3. **Placebo independence.** The one-sided lower bound of $\text{IR}(4) - \text{IR}(2)$ exceeds zero — a comparison that does not involve the placebo at all, so a placebo pathology cannot by itself produce a positive verdict.
4. **Economic materiality.** $\Delta\text{IR}(4 - 5) \geq 0.5$ annualized information ratio, fixed before data. The rule string names the tested quantity as (information) + (capacity handicap), so the manifest cannot be read as claiming the information effect alone cleared it.
5. **Multiplicity.** The deflated Sharpe ratio is at least 0.95 over the enumerated trial set.

Two guards sit outside the clauses, and both are HALT-only: they can refuse to report an outcome their own inputs cannot support, and they can never turn SURVIVES into NULL or the reverse. The degeneracy guard fails closed if any cell-seed pair yielded non-finite forecast diagnostics, or if a required diagnostic is missing — a response to a defect in which the guard reported itself armed while examining nothing. The power guard compares each cell’s across-seed range to the effect being claimed and halts if the wobble of the inputs reaches the size of the claim.

The outcome taxonomy has four words and all four are publishable: SURVIVES, NULL, INCONCLUSIVE, and HALT_ADJUDICATE. Committing to INCONCLUSIVE in advance is what makes the design honest under low power — the response to an underpowered result was written out and signed before the seed count was fixed, so “we did not have the power” is a pre-registered outcome rather than a post-hoc excuse.

Two further requirements are enforced at the manifest level rather than trusted to prose. A verdict manifest missing any required field cannot be quoted, and the loader refuses it; the required set includes a `dual_specification` field carrying the verdict computed under both the current and the pre-amendment specifications from the same artifacts, with any disagreement between them a first-class finding rather than a footnote. And every manifest carries a required disclosure, refused if absent: *we cannot exclude that our BSQ baseline is weaker than a reference BSQ implementation, which would inflate the FSQ-versus-BSQ comparison.*

4.7 Multiple testing and power

The deflated Sharpe ratio corrects for the number of configurations the design could have reported, and that number is enumerated rather than estimated: five cells $\times$ three horizons $\times$ four values of $\kappa = 60$ trials. The factorization is computed from named axes in code and asserted by a gate, together with the negative form proving the count does not silently acquire a seeds axis.

Seeds do not enter that count, and the reason is worth stating precisely because it looks like a convenience. The five seeds are *replicates of one configuration*, not five configurations we could have chosen to report. A multiple-testing adjustment cannot legitimately become stricter because the same specification was repeated more times; if it did, replication — which reduces variance — would be penalized as though it were search. The full 300-value cross-product of cells, seeds, horizons and κ is retained as an audit trail so that a reader can verify no seed-level selection occurred, but the correction is applied over 60.

The null dispersion the deflation uses is taken from the placebo cell alone. The earlier specification pooled all arms, which made this clause anti-correlated with its own hypothesis: a larger treatment effect inflated the pooled dispersion and made the clause harder to pass, to the point of requiring a cell-4 information ratio of 3–10 against an economic floor of 0.5. A proposed fallback to cell 2 was rejected because §5 can claim $IR(2) - IR(1)$, which makes cell 2 a treatment arm rather than a null. The amendment was made before any data and its direction was recorded in advance as indeterminate: it loosens clause 5 by 0.859 $\times$ while clause 2 tightens by 1.24–1.60 $\times$.

The minimum detectable effect carries two terms — scoring noise and training variance — and is computed with a self-written Student- t quantile and Welch–Satterthwaite degrees of freedom rather than a normal approximation, because at three to five seeds the degrees of freedom are small enough that the difference matters. The reference figure at the pinned horizon is 3.518 annualized information ratio.

We state the consequence plainly, in advance. That reference figure stands against an economic floor of 0.5 and a prior, measured on this data at an earlier milestone, of $|RankIC| \approx 0.027$ for trade-flow imbalance. The paired minimum detectable effect governs and will sit below the reference figure to the extent that cells 4 and 5 share exposure, by an amount not knowable before the data. **On these numbers, NULL or INCONCLUSIVE is the most likely honest outcome**, and nothing in the design manufactures power it does not have.

4.8 Pre-registration as the integrity mechanism

Everything above — the cells, the parity band, the placebo construction, the five clauses and their thresholds, the horizon, the cost, the κ grid, the seed count, the trial count and the null basis — is pinned on a conformance surface that the runner checks before it trains anything, and each pin carries a mutation test proving the gate rejects its predecessor value. Changing a pin is not a code edit that a reviewer must catch; it fails the run.

The pre-registration is a dated document with an append-only amendment log. Every amendment records what changed, when, why, and — where the direction was knowable — whether it made the bar easier or harder, recorded before the data that would reveal which. Amendments made after seeing calibration results are marked as such, including the one described in Section 3.6. The log also contains withdrawn claims: statements this project made and later retracted, retained rather than deleted. We regard that as the load-bearing part. A pre-registration that only ever records successful predictions is not evidence of discipline; one that records its own corrections is.

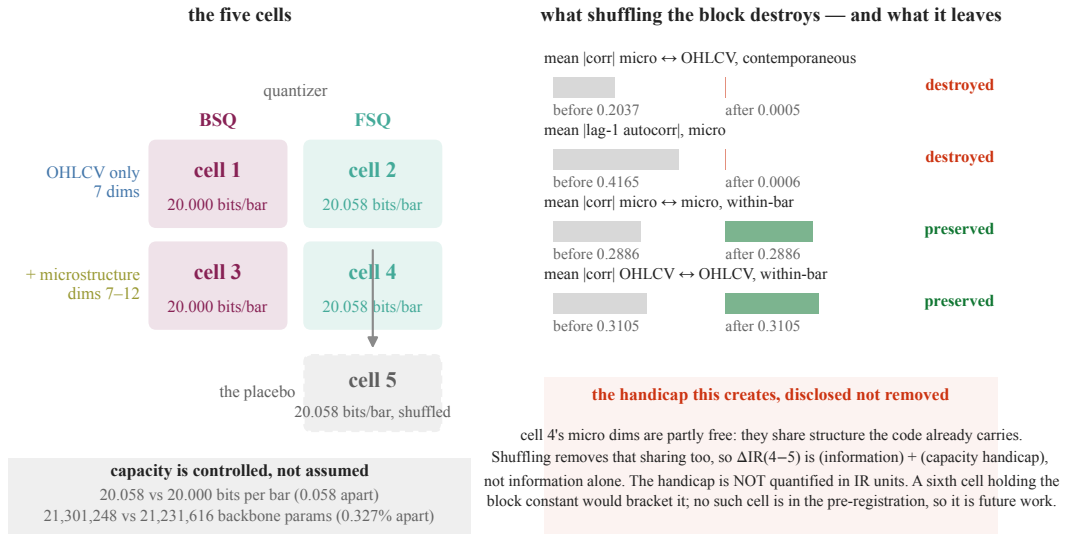


Figure 5: The five cells, at matched capacity, and the exact reach of the placebo. *Left:* a 2×2 over quantizer and input arm, plus a fifth cell that is cell 4 with the microstructure block shuffled. Capacity is controlled rather than assumed: the two quantizer arms differ by 0.058 bits per bar and by 0.327% of backbone parameters, so a difference between them cannot be attributed to one arm simply having more room. *Right:* what the shuffle actually does, measured. It destroys the two things it was built to destroy — microstructure's contemporaneous link to price ($0.2037 \rightarrow 0.0005$) and its own lag-1 autocorrelation ($0.4165 \rightarrow 0.0006$) — and leaves the within-block and within-OHLCV correlations exactly as they were. The consequence is stated rather than hidden: because cell 4's microstructure dimensions are partly amortized against structure the code already carries, shuffling removes that sharing too, so $\Delta IR(4-5)$ is (information) + (capacity handicap). The handicap is not quantified in information-ratio units. A sixth cell holding the block constant would bracket it, but no such cell appears anywhere in the pre-registration, so it is future work rather than a committed contingency, and Section 7 carries the handicap as an open disclosure.

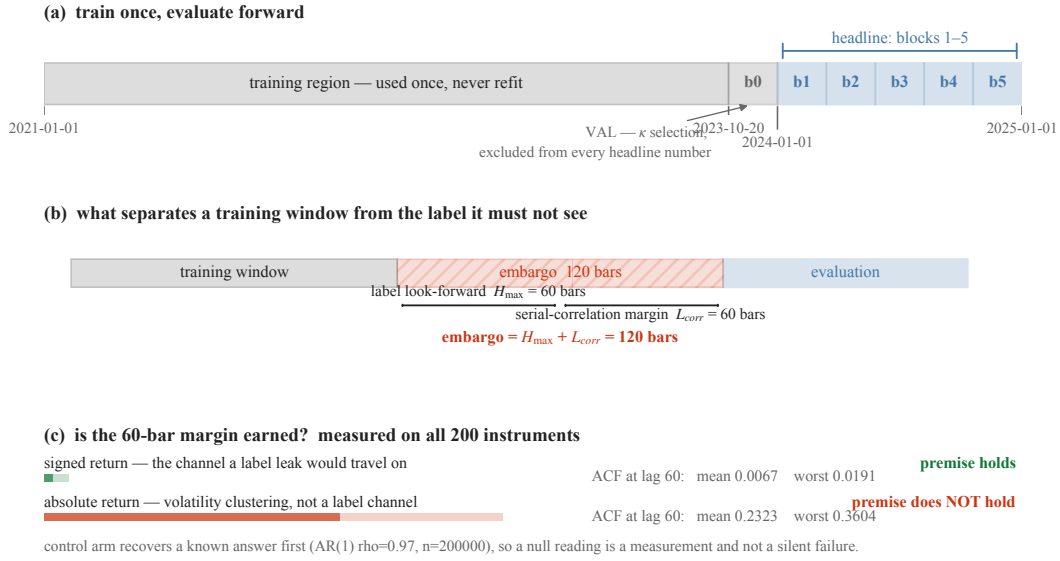


Figure 6: Train once, evaluate forward, and pay for the separation. (a) The training region ends at one boundary and is never refit. The forward region is cut into six blocks; block 0 is spent selecting the execution filter and is excluded from every headline number, so the headline is scored on five blocks the selection never saw. (b) Between a training window and any label that could contaminate it sit a 60-bar label look-forward and a 60-bar serial-correlation margin, giving a 120-bar embargo. (c) The premise behind that margin is measured on all 200 instruments rather than assumed, and a control arm recovers a known AR(1) answer first, so a null reading is a measurement and not a silent failure. It holds for signed returns — the channel a label leak would travel on — at ACF 0.0067 mean and 0.0191 worst-symbol at lag 60. It does *not* hold for absolute returns (0.2323 mean, 0.3604 worst), which is volatility clustering rather than a label channel; Section 7 carries that caveat.

References